Calculați $F(a) = \int_0^{\pi/2} \ln(a^2 - \sin^2\theta)\, d\theta$, $a > 1$

$$F'(a) = \int_0^{\frac{\pi}{2}} \frac{2a}{a^2 - \sin^2\theta}\, d\theta = \frac{\pi}{\sqrt{a^2-1}} \quad (\text{vezi Curs 20})$$

$\exists\, C \in \mathbb{R}$ a.î. $F(a) = \pi \ln\left(a + \sqrt{a^2-1}\right) + C$

$$F(a) = \int_0^{\frac{\pi}{2}} \ln\left[a^2\left(1 - \frac{\sin^2\theta}{a^2}\right)\right] d\theta = \int_0^{\frac{\pi}{2}} 2\ln a\, d\theta + \int_0^{\frac{\pi}{2}} \ln\left(1 - \frac{\sin^2\theta}{a^2}\right) d\theta$$

$$= \pi \ln a + \int_0^{\frac{\pi}{2}} \ln\left(1 - \frac{\sin^2\theta}{a^2}\right) d\theta$$

$$1-\frac{\sin^2\theta}{a^2} \geqslant 1-\frac{1}{a^2} \Rightarrow 0 > \ln\left(1-\frac{\sin^2\theta}{a^2}\right) \geqslant \ln\left(1-\frac{1}{a^2}\right)$$

$$\Rightarrow \left|\ln\left(1-\frac{\sin^2\theta}{a^2}\right)\right| \leq \left|\ln\left(1-\frac{1}{a^2}\right)\right| \quad , \forall a > 1$$

$$\left|\int_0^{\frac{\pi}{2}} \ln\left(1-\frac{\sin^2\theta}{a^2}\right) d\theta\right| \leq \int_0^{\frac{\pi}{2}} \left|\ln\left(1-\frac{\sin^2\theta}{a^2}\right)\right| d\theta$$

$$\leq \int_0^{\frac{\pi}{2}} \left|\ln\left(1-\frac{1}{a^2}\right)\right| d\theta \leq \frac{\pi}{2}\left|\ln\left(1-\frac{1}{a^2}\right)\right| \xrightarrow{a\to\infty} 0 .$$

$$\lim_{a\to\infty} = F(a) - \pi \ln a = \lim_{a\to\infty} \int_0^{\frac{\pi}{2}} \ln\left(1 - \frac{\sin^2\theta}{a^2}\right) d\theta = 0.$$

$$\|$$

$$\lim_{a\to\infty} \pi \ln\left(a + \sqrt{a^2-1}\right) + C - \pi \ln a$$

$$\|$$

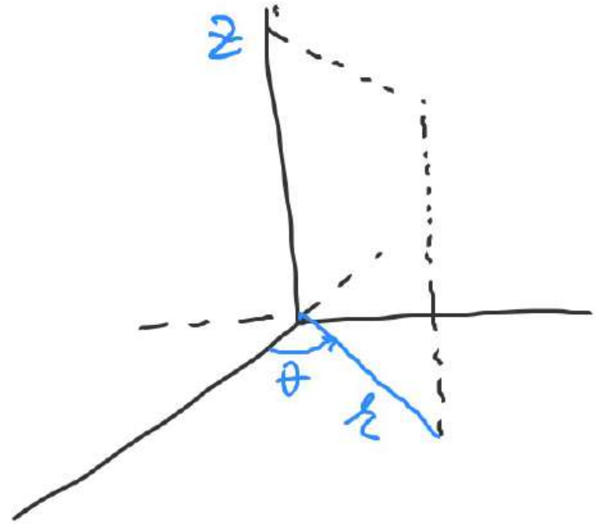
$$\lim_{a\to\infty} \pi \ln\left(\frac{a+\sqrt{a^2-1}}{a}\right) + C$$

$$\|$$

$$\pi \ln 2 + C = 0 \Longrightarrow C = -\pi \ln 2.$$

Deci $\quad F(a) = \pi \ln \dfrac{a+\sqrt{a^2-1}}{2}$

Coordonate Cilindrice



$$\phi : [0, \infty] \times [0, 2\pi] \times \mathbb{R} \longrightarrow \mathbb{R}^3$$

$$\phi(r, \theta, z) = (r\cos\theta, r\sin\theta, z)$$

$$(0, \infty) \times (0, 2\pi) \times \mathbb{R} \xleftrightarrow{\phi} \mathbb{R}^3 \setminus \{(x, 0, z),\ x \geq 0\}$$

difeomorfism de clasă C

$$\det J_\phi(r, \theta, z) = \begin{vmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = r$$

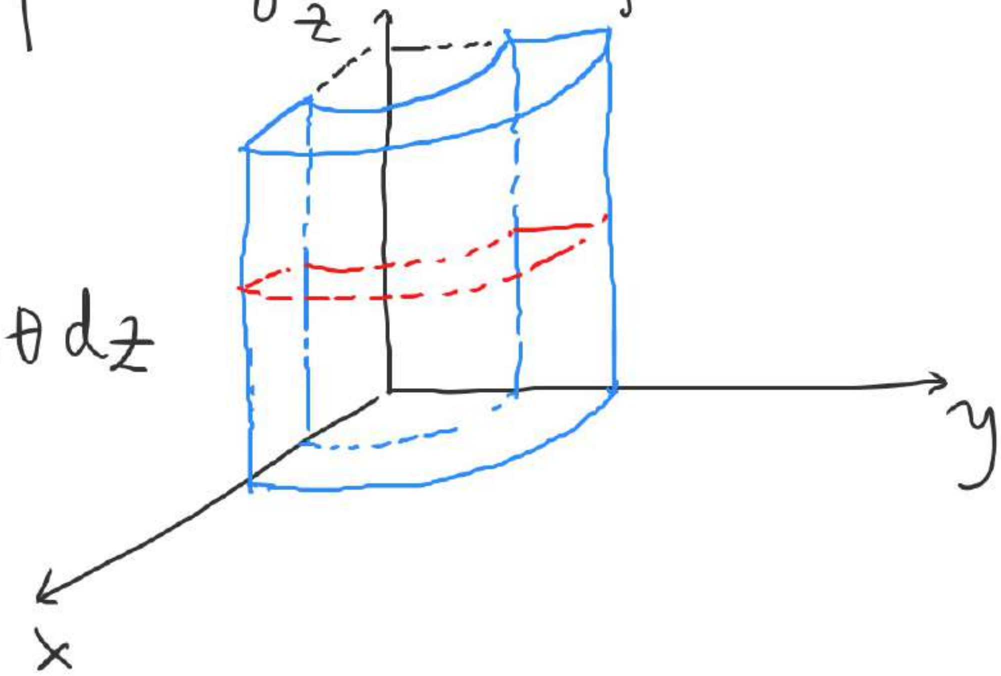
$$\det J_\phi(r, \theta, z) = r$$

$A \in J(\mathbb{R}^3)$, $f : A \to \mathbb{R}$ integrabilă

$$\iiint_A f(x,y,z)\,dx\,dy\,dz = \iiint_{\phi^{-1}(A)} f(r\cos\theta,\, r\sin\theta,\, z)\, \overset{r.}{\overset{\|}{\left|\det J_\phi\right|}}\, dr\,d\theta\,dz$$

$$dx\,dy\,dz = \left|\det J_\phi(r,\theta,z)\right| dr\,d\theta\,dz = r\,dr\,d\theta\,dz$$

$$I = \iiint_V xz\,dx\,dy\,dz \qquad V = \{(x,y,z) \mid 4 \le x^2+y^2 \le 9,\ x \geq 0,\ y \geq 0,\ 0 \le z \le 2\}$$



$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ z = z \end{cases} \qquad d\ dy\,dz = r\,dr\,d\theta\,dz$$

$$(x,y,z) \in V \iff \begin{cases} r \in [2,3] \\ \theta \in [0, \pi/2] \\ z \in [0,2] \end{cases}$$

$$I = \int_2^3 \left(\int_0^{\pi/2} \left(\int_0^2 r\cos\theta \cdot z \cdot r\,dz \right) d\theta \right) dr$$

$$= \int_2^3 r^2\,dr \cdot \int_0^{\frac{\pi}{2}} \cos\theta\, d\theta \cdot \int_0^2 z\, dz = \frac{19}{3} \cdot 1 \cdot 2 = \frac{38}{3}$$

sau:

$$D = pr_{xOy} V = \{ (x,y) \in \mathbb{R}^2 \mid 4 \le x^2+y^2 \le 9,\ x,y \ge 0 \}$$

$$I = \iint_D \left(\int_0^2 xz\, dz \right) dx\,dy = \iint_D 2x\,dx\,dy = \dots = \frac{38}{3}$$

sau

$$I = \int_0^2 \left(\iint_{V_z} xz\,dx\,dy \right) dz = \int_0^2 \left(\iint_D xz\,dx\,dy \right) dz = \dots$$

$A = \{(x,y,z) \mid x^2+y^2+z^2=4, \ z \geq 0\}$

Aratati că $A \in \mathcal{J}(\mathbb{R}^3)$ și $\lambda(A)=0$.

$(x,y,z) \in A \Longleftrightarrow \left(z = \sqrt{4-x^2-y^2}, \ (x,y) \in D = \{(x,y) \mid x^2+y^2 \leq 4\} \right)$

$(D = pr_{xOy} A)$.

· $D \in \mathcal{J}(\mathbb{R}^2)$

$f : D \to \mathbb{R}$, $f(x,y) = \sqrt{4-x^2-y^2}$ integr. Riemann

pt că f cont și mărg

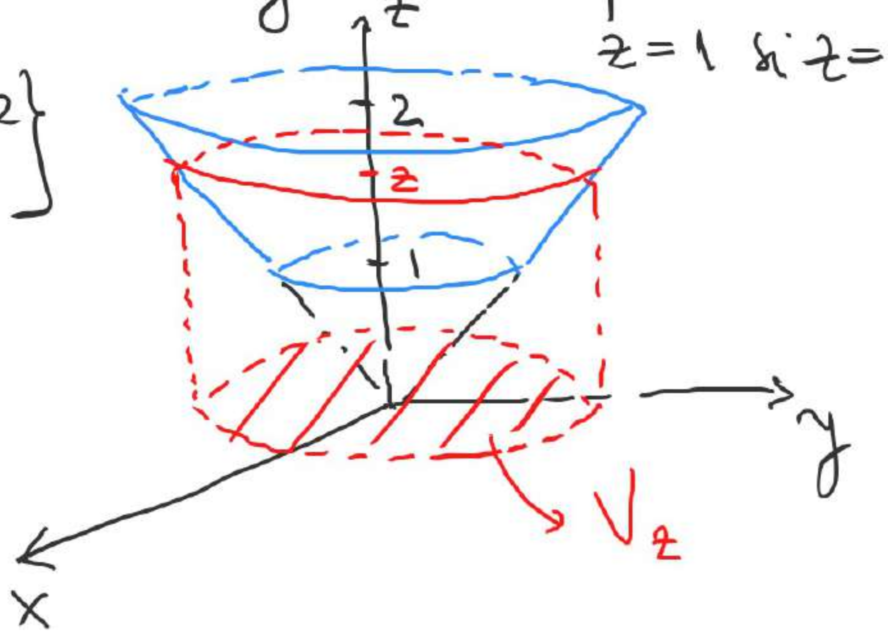
$\Rightarrow A = G_f \in \mathcal{J}(\mathbb{R}^3)$ și $\lambda(A) = 0$ (Curs 17, pag 4-5).

$B = \{(x,y,1) \mid x^2+y^2 \leq 4\} \subset [-2,2] \times [-2,2] \times \{1\} = A$

$\left.\begin{array}{l} \lambda(A) = 0 \\ B \subset A \end{array}\right\} \Longrightarrow B \in \mathcal{J}(\mathbb{R}^3)$ şi $\lambda(B) = 0$.

Calculaţi $J = \iiint\limits_V z^2 \, dx\,dy\,dz$

unde V este mărginit de conul $x^2+y^2=z^2$ şi planele $z=1$ şi $z=2$



$V_z = \{(x,y) \mid x,y,z \in V\} = \{(x,y) \mid x^2+y^2 \leq z^2\}$

$J = \int_1^2 \left(\iint\limits_{V_z} z^2 \, dx\,dy \right) dz$

$$\iint_{V_z} z^2\,dx\,dy = z^2 \iint_{V_z} dx\,dy = z^2 \lambda(V_z) = z^2\cdot \pi z^2 = \pi z^4$$

$$J = \int_1^2 \pi z^4\,dz = \frac{\pi z^5}{5}\Big|_1^2 = \frac{\pi}{5}\cdot 31$$

Metoda 2. (cu coord. cilindrice)

$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ z = z \end{cases} \qquad (x,y,z)\in V \iff \begin{cases} r\in[0,z] \\ \theta\in[0,2\pi] \\ z\in[1,2] \end{cases}$$

sau $\begin{cases} r\in[0,2] \\ \theta\in[0,2\pi] \\ z\in[1,2] \text{ pt } r\le 1 \\ z\in[r,2] \text{ pt } r\ge 1 \end{cases}$

$$J = \int_1^2\left(\int_0^{2\pi}\left(\int_0^z z^2\cdot \underline{r\,dr}\right)d\theta\right)dz = \int_1^2\left(\int_0^{2\pi} z^2\cdot\frac{z^2}{2}\,d\theta\right)dz$$

$$= \int_1^2 \pi z^4 \, dz = 31 \cdot \frac{\pi}{5}$$

$$V \longleftrightarrow V' = \{(r,\theta,z) \mid r \in [0,1], \theta \in [0,2\pi], z \in [1,2]\} \cup$$

$$\cup \{(r,\theta,z) \mid r \in [1,2], \theta \in [0,2\pi], z \in [r,2]\}$$

$$J = \int_0^1 r \, dr \cdot \int_0^{2\pi} d\theta \cdot \int_1^2 z^2 dz + \int_0^{2\pi} \left(\int_1^2 \left(\int_r^2 z^2 \cdot r \, dz \right) dr \right) d\theta$$

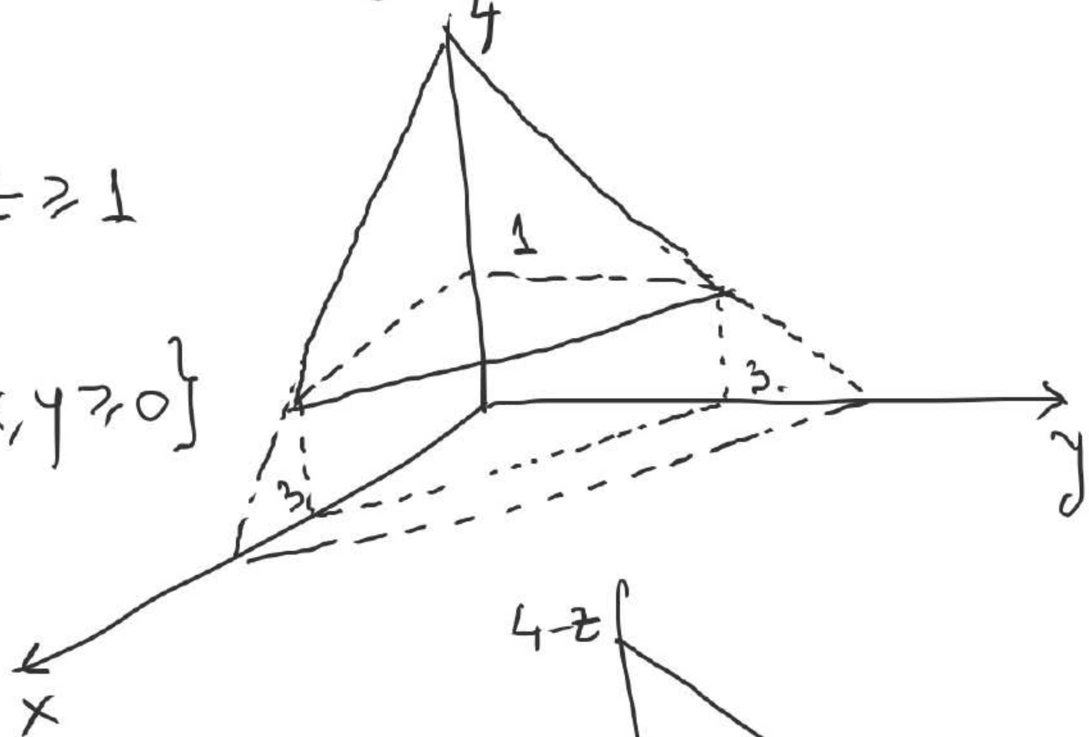
$$= 2\pi \cdot \frac{1}{2} \cdot \frac{7}{3} + 2\pi \cdot \int_1^2 \frac{z^3}{3} \cdot r \Big|_{z=r}^{z=2} dr = \frac{7\pi}{3} + 2\pi \int_1^2 \frac{8-r^3}{3} \cdot r \, dr$$

$$= \frac{7\pi}{3} + \frac{2\pi}{3}\left(4r^2\Big|_1^2 - \frac{r^5}{5}\Big|_1^2\right) = \frac{7\pi}{3} + \frac{2\pi}{3}\left(12 - \frac{31}{5}\right) = \frac{35\pi}{15} + \frac{2\pi}{3} \cdot \frac{29}{5} = \frac{31}{5}\pi.$$

$$I=\iiint_V z\,dx\,dy\,dz$$

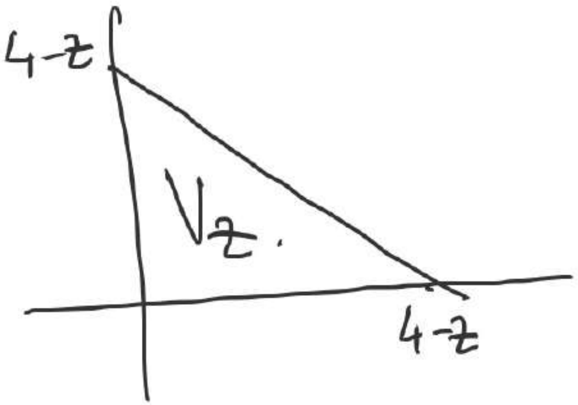
V este mărginit de planele

$x+y+z=4$, $x=0$, $y=0$ și $z=1$



$V: x+y+z \le 4,\ x \ge 0,\ y \ge 0,\ z \ge 1$

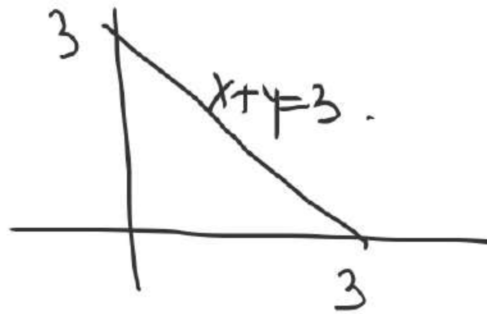
1 $V_z = \{(x,y) \mid x+y \le 4-z,\ x,y \ge 0\}$



$$I=\int_1^4 \left(\iint_{V_z} z\,dx\,dy\right) dz = \int_1^4 z\,\lambda(V_z)\,dx\,dy$$

$$= \int_1^4 z \cdot \frac{(4-z)^2}{2}\, dz = \dots$$

② $pr_{xOy} V = D = \{ (x,y) \in \mathbb{R}^2 \mid x+y \leq 3,\ x, y \geq 3 \}$



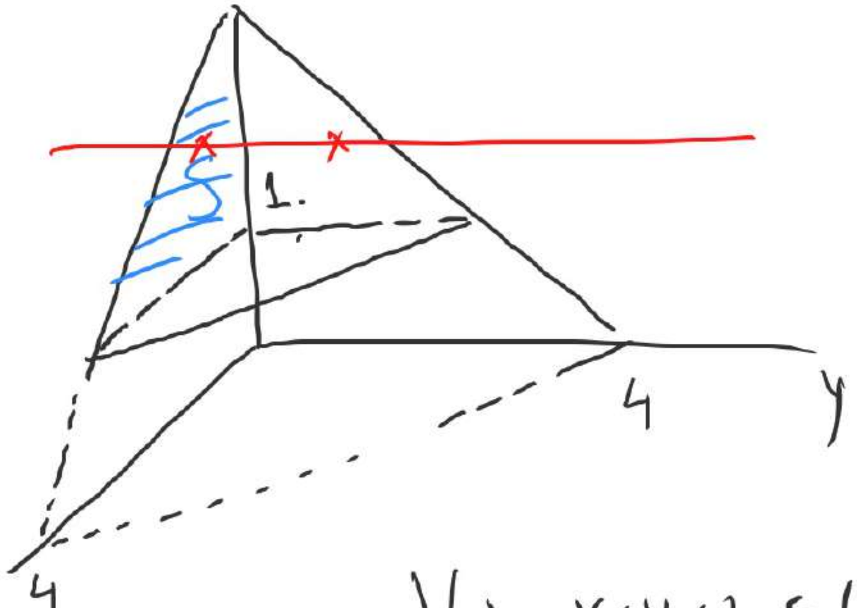
$$V = \{ (x,y,z) \mid (x,y) \in D,\ 1 \leq z \leq 4-x-y \}$$

$$I = \iint_D \left(\int_1^{4-x-y} z\, dz \right) dx\,dy = \int_0^3 \left(\int_0^{3-x} \left(\int_1^{4-x-y} z\, dz \right) dy \right) dx$$

$$= \int_0^3 \left(\int_0^{3-x} \frac{(4-x-y)^2 - 1}{2}\, dy \right) dx$$

$$S = pr_{xOz} V$$

$$S = \{(x,z) \mid x+z \le 4,\ z \ge 1,\ x \ge 0\}$$



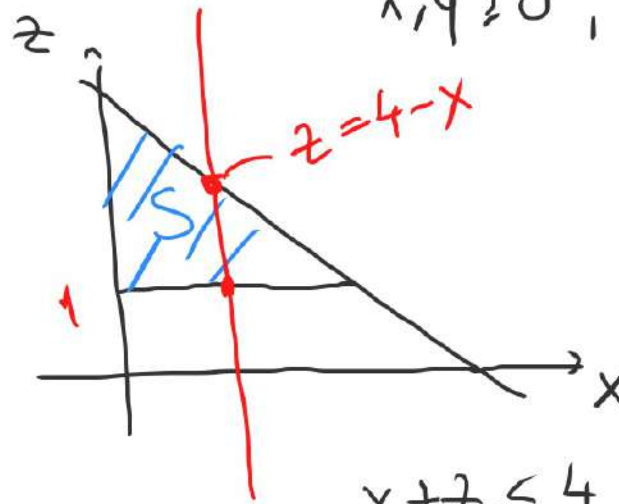
$$V = \{(x,y,z) \mid (x,z) \in S,\ 0 \le y \le 4-x-z\}$$

$V:\ x+y+z \le 4$

$x, y \ge 0,\ z \ge 1.$

$$J = \iint_S \left(\int_0^{4-x-z} z\, dy \right) dx\, dz$$

$$= \iint_S z(4-x-z)\, dx\, dz =$$



$S:\ x+z \le 4$

$z \ge 1,\ x \ge 0.$

$S:\ \{(x,z) \mid 0 \le x \le 3,\ 1 \le z \le 4-x\}$

$$= \int_0^3 \left(\int_1^{4-x} z(4-x-z)\, dz \right) dx$$

$$= \int_0^3 \left(\int_1^{4-x} \left[z(4-x) - z^2 \right] dz \right) dx = \int_0^3 \left((4-x) \cdot \frac{z^2}{2} \Big|_1^{4-x} - \frac{z^3}{3} \Big|_1^{4-x} \right) dx$$

$$= \int_0^3 \left[\frac{(4-x)^3}{2} - \frac{(4-x)^3}{3} - \frac{4-x}{2} + \frac{1}{3} \right] dx = \int_0^3 \left(-\frac{(x-4)^3}{6} + \frac{x-4}{2} + \frac{1}{3} \right) dx$$

$$= -\frac{(x-4)^4}{24} \Big|_0^3 + \frac{(x-4)^2}{4} \Big|_0^3 + 1 = -\frac{1}{24} + \frac{256}{24} + \frac{1}{4} - 4 + 1 = \frac{255+6-72}{24} = \frac{63}{8}$$